

OpenQT

Multilingual DOS Word Processor

Version 3.8.0 - Author: Ronen Blumberg - Public Domain

OpenQT User Manual

A Hebrew / Arabic / Russian / English word processor for DOS

Hebrew - Arabic - Russian - English

True trilingual editing with full bidirectional (RTL) support

Inline formatting, undo/redo, encryption, UTF-8 round-trip

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1. Introduction

OpenQT is a full-screen word processor for DOS that writes Hebrew, Arabic, Russian and English. It is a modern clone of the classic QText 5.5, rebuilt in C with full bidirectional (right-to-left) support, inline formatting, undo/redo, password encryption, on-screen Arabic cursive shaping, and -- uniquely -- true trilingual editing in which Hebrew, Arabic and English all coexist on the same line of the same document, simultaneously visible on the VGA screen.

The editor draws directly to VGA text-mode memory and uses BIOS interrupts for input, so it has no dependencies beyond plain DOS. It runs in real-mode 16-bit DOS or, preferably, 32-bit via the bundled DOS4GW extender. Inside DOSBox or DOSBox-X it runs on any modern Linux, Windows or macOS host.

Author: Ronen Blumberg. License: Public Domain. Version: OpenQT 3.9.0.

Key features

- Full BiDi: Hebrew and Arabic right-to-left, English and digits left-to-right, freely interleaved.
- Inline bold and underline formatting.
- Up to 30,000 lines per document; undo/redo 250 levels deep.
- Search and search-and-replace across all scripts.
- Block (range) operations: copy, cut, paste, delete, reformat.
- Word wrap and paragraph reflow to column 71.
- Password protection with the OQT-ENC1 encrypted format.
- On-screen CP864 Arabic positional shaping.
- Mouse support (clickable menus, click-to-caret, drag-select, wheel).
- Host-assisted tools: spell check, text-to-speech, dictation, machine translation and Unicode clipboard paste.

2. Quick start

The compiled editor and all tools ship ready to run. Mount the OpenQT folder as C:\OPENQT inside DOSBox (or DOSBox-X):

```
mount c /path/to/openqt
c:
cd \OPENQT
```

Then launch the editor in the language combination you want:

```
OQT 3 mydoc.txt      # Hebrew + Arabic + English (trilingual)
OQTH mydoc.txt      # Hebrew + English
OQTA mydoc.txt      # Arabic + English
OQTR mydoc.txt      # Russian + English (CP866)
```

The first few keys to know:

Key	Action
F4	Cycle input language: English -> Hebrew -> Arabic -> Russian
F5	Toggle right-to-left layout
F2	Save
F3	Open another file
F1	Quick help screen
Alt+H	Help menu -> Docs... opens the in-editor guide
Alt+X	Exit

3. Launchers

Each launcher is a small BAT file that prepares the VGA character-generator font for its language set, then starts the editor with the right flags. Always launch through these rather than running OPENQT.EXE directly, so the screen font matches your text.

Launcher	What it sets up
OQT 3 file	Trilingual: HEBVGA, then ARABVGA /P overlay, then OPENQT /A. Hebrew + Arabic + English on one screen.
OQTH file	Hebrew + English. Runs HEBVGA, then OPENQT.
OQTA file	Arabic + English. Runs ARABVGA, then OPENQT /A.
OQTR file	Russian + English (CP866). Runs RUSVGA, then OPENQT /R.
OQT file	Default -- same as OQTH (Hebrew + English).

The /A flag tells OpenQT that Arabic glyphs occupy the 0xC0-0xDF byte range where CP437 keeps its box-drawing characters, so it draws dialog frames with plain ASCII (+ - |) instead. Russian (/R) keeps CP437 box-drawing intact and needs no such fallback.

4. Keyboard reference

Files and editing

Key	Action
F1	Help screen
F2	Save current file
F3	Open another file (file dialog)
Ctrl+S	Save without a dialog
Ctrl+N	New document
Ctrl+Z	Undo
Ctrl+Y	Redo
Insert	Toggle insert / overwrite
Delete	Delete character under cursor
Backspace	Delete character to the left

Cursor movement

Key	Action
Arrow keys	Move one character / one line
Ctrl+Left / Ctrl+Right	Move five characters at a time
Home / End	Start / end of line
PgUp / PgDn	Page up / down
Ctrl+Home / Ctrl+End	Start / end of document
Ctrl+G	Go to line number

Language and direction

Key	Action
F4	Cycle input language: English -> Hebrew -> Arabic -> Russian (auto-flips RTL/LTR to match)
F5	Toggle right-to-left paragraph direction
F10	Toggle 'embedded LTR' -- type English/digits inline inside an RTL paragraph without changing language

Search, formatting, blocks

Key	Action
F7	Find
F8	Replace
Ctrl+B	Toggle bold for the next typed text
Ctrl+U	Toggle underline
Alt+U	Toggle bold + underline together
F9	Paste block at cursor
Alt+V	Paste host clipboard (via helper)
Alt+C	Copy selection/document to host clipboard
Alt+B	Open Block menu (start/end/copy/cut/paste/delete/reformat)

Menus

Key	Menu
Alt+F	File
Alt+E	Edit
Alt+S	Search
Alt+B	Block
Alt+O	Options
Alt+T	Tools (host-assisted features)
Alt+H	Help (Help / Docs... / About...)
Alt+X	Exit

5. Typing layouts

OpenQT maps incoming ASCII keystrokes to letters itself, so set your DOSBox keyboard to a plain US layout. F4 chooses which script the next keystrokes produce.

Hebrew (standard Israeli layout)

Key	Letter	Key	Letter	Key	Letter
t	א	c	ב	d	ג
s	ד	v	ה	u	ו
z	ז	j	ח	y	ט
h	י	l	ך כ	f	ל
k	ם מ	n	ן נ	b	ס
g	ע	;	ף פ	m	ץ צ
e	ק	r	ר	a	ש
,	ת				

Final-letter (sofit) substitution is automatic at the end of a word.

Arabic (Microsoft Arabic 101 layout)

Key	Letter	Key	Letter	Key	Letter
a	ش	b	لآ	c	ف
d	ي	e	ن	f	ب
g	ج	h	ا	i	ه
j	ت	k	ن	l	م
m	ة	n	س	o	خ
p	ح	q	ض	r	ق
s	س	t	ف	u	ع
v	ر	w	ص	x	ء
y	غ	z	ئ	[	ج
]	د	`	ذ		

Digits 0-9 give Arabic-Indic numerals. Punctuation keys map to extra letters: ' -> Tah, , -> Waw, ; -> Kaf, . -> Zain, / -> Zah.

Russian (JCUKEN layout)

Key	Letter	Key	Letter	Key	Letter
q	й	w	ц	e	у
r	к	t	е	y	н
u	г	i	ш	o	щ
p	з	[	х	]	ъ
a	ф	s	ы	d	в
f	а	g	п	h	р
j	о	k	л	l	д
;	ж	'	э	z	я
x	ч	c	с	v	м
b	и	n	т	m	ь
,	б	.	ю	`	ё

Russian has letter case (Shift gives uppercase). Digits 0-9 pass through; / gives a period and ? a comma (JCUKEN swap).

6. Trilingual mode

OQT 3 lets Hebrew, Arabic and English share the same line of the same document, all visible at once. The trick is byte ranges: the CP862 Hebrew font puts letters at 0x80-0x9A while the CP864 Arabic font puts letters at 0xA0-0xFE. These ranges do not overlap, so both fonts can live in the VGA character generator at the same time.

The launcher loads a full CP862 base font with HEBVGA, then overlays only the 95 Arabic slots with ARABVGA /P. The result is a single screen font carrying all three scripts. Press F4 to choose which range your keystrokes land in; the BiDi engine arranges direction automatically.

```
F4 -> English   type:  See the word
F4 -> Hebrew   type:  שלום
F4 -> Arabic    type:  مرحبا
F4 -> English   type:  in three scripts.
```

A saved trilingual file is a mixed CP862+CP864 byte stream -- not a standard codepage. To share it outside OpenQT, export to UTF-8 with oqt2word (see chapter 13).

7. Russian mode

Russian (CP866) runs as its own session -- English + Russian only. It cannot mix with Hebrew or Arabic because CP866 Cyrillic overlaps both the CP862 Hebrew range (0x80-0x9F) and the CP864 Arabic range (0xA0-0xAF, 0xE0-0xEF). The same byte slot would have to be a Hebrew, an Arabic and a Cyrillic letter at once, which 8-bit storage cannot express.

Launch with OQTR. F4 still cycles through all four languages, and the direction auto-flips: RTL for Hebrew/Arabic, LTR for English/Russian. Everything else -- bold, underline, undo, search, blocks, encryption -- behaves exactly as in English mode. Because CP866 keeps CP437 box-drawing intact, dialogs render normally with no /A fallback.

A document written in Russian mode is a standard CP866 file. If reopened in Hebrew/Arabic mode its letters render as the wrong script -- the 8-bit storage carries no language tag. Use the converter /R flag to move Russian text to/from UTF-8.

8. BiDi and Arabic shaping

The file always stores text in logical order -- the order you typed the letters. At display time OpenQT reorders the bytes to their visual position: it strips inline format codes, reverses the line when the paragraph is RTL, then re-reverses each English/digit run so numbers and Latin words read left-to-right within the larger RTL flow. The cursor maps keep the caret correct even when visual order differs from typing order.

Arabic positional shaping

OpenQT shapes Arabic cursively on the DOS screen: each letter is drawn in its correct positional form (isolated, initial, medial, final) based on its neighbours, so the text joins like real Arabic. This is display-only -- the file always stores the abstract isolated form, so saving, loading and the converters are unaffected.

A few CP864 letters (Tah, Zah, Yeh-Hamza, and plain Lam-Alef without madda) lack separate positional glyphs and appear in their isolated shape inside a join -- a small on-screen imperfection. Exported UTF-8 is always correct because the receiving program does its own shaping.

9. Formatting, blocks and reflow

Bold and underline

Press Ctrl+B to toggle bold and Ctrl+U to toggle underline for the text you type next; Alt+U toggles

both together. Formatting is stored as in-band toggle bytes inside the line and resets at the start of each line.

Block operations

Open the Block menu with Alt+B. Mark a block (start, move, end), then copy, cut, paste (also F9), or delete it. Blocks use the same coordinates as mouse drag-selection, so the two are interchangeable.

Reformat (reflow)

Block -> Reformat re-wraps text to the column-71 margin: short lines are joined and long lines are re-split at word boundaries, while blank lines are preserved as paragraph breaks. With nothing marked it reflows the whole document; with a block marked it reflows only those lines. It works on logical-order bytes (BiDi-safe) and keeps bold/underline travelling with their words.

Reformat rebuilds the buffer in one pass and cannot be undone. It asks you to confirm first and clears the undo history -- save before running it if you want a fallback.

10. Mouse

If a DOS mouse driver is present (DOSBox and DOSBox-X provide one), OpenQT becomes clickable. With no driver the mouse is simply inert and every keyboard command still works.

Action	Result
Click a top-bar title	Open that menu
Click a menu item	Run it
Click in the text	Move the cursor there (LTR and RTL)
Press and drag	Select a block
Drag past top/bottom	Auto-scroll while selecting
Mouse wheel	Scroll (three lines per notch)

11. In-editor Docs viewer

Alt+H -> Docs... opens a paged, read-only guide over the editor without disturbing your document (Esc returns exactly where you were). The viewer loads one of four .HLP files chosen by the current input language, so the loaded VGA font always matches the file's content:

Active language	File loaded
Hebrew	OPENQTH.HLP (Hebrew + English)
Arabic	OPENQTA.HLP (Arabic + English)
Russian	OPENQTR.HLP (Russian + English)
English / default	OPENQT.HLP (trilingual)

Each line is rendered through the same BiDi and Arabic-shaping pipeline as the editor. Page with Up/Down, PgUp/PgDn, Home/End; close with Esc or Q.

12. Host-assisted tools

The Tools menu (Alt+T) hands work to a small helper daemon running on the host (Linux, Windows or macOS), because DOS has no microphone, no speech engine, no modern network, and no UTF-8 clipboard. Start the helper on the host first (host_helper/run_helper.sh); if it is not running these items simply report 'Helper not responding' and the rest of the editor is unaffected.

Tools item	What it does	Engine
Spell Check	Flags unknown words with numbered suggestions; follows the F4 language (English/Hebrew)	aspell/hspell
Read Aloud	Speaks the selection or document; voice follows F4 language	espeak-ng
Translate to Hebrew	Translates English and inserts Hebrew (needs internet)	online MT
Dictate	Records the mic and inserts recognised text (Eng/Heb)	whisper
Paste Host	Inserts host clipboard as clean text for the F4 language	xclip

Copy to Host	Sends the selection or document to the host clipboard as UTF-8 (Alt+C)	xclip
--------------	--	-------

Paste Host (also Alt+V) strips niqqud and harakat and folds smart quotes and dashes to ASCII -- use it instead of DOSBox-X's Ctrl+F6, which mangles RTL text by re-mapping it through the DOS keyboard layout.

13. File format and encryption

OpenQT files are plain DOS text files (CR/LF line endings) in a mixed byte encoding. Each byte range maps to one script:

Byte range	Meaning
0x00-0x7F	ASCII (English, digits, punctuation)
0x80-0x9A	Hebrew letters (CP862)
0xA0-0xFE	Arabic letters (CP864 isolated form)
0x02 / 0x03 / 0x04	Bold / underline / both toggles
0xAF / 0xAE	QText 5.5 markers (recognised on read)

Russian documents are instead standard CP866 byte streams; their ranges overlap Hebrew and Arabic, which is why CP866 cannot share a file with them.

Encryption

Saving with a password prefixes the file with the magic header OQT-ENC1 and encrypts the rest byte-by-byte; OpenQT detects the header on load and prompts for the password. The cipher is a deterrent against casual reading, not serious cryptography -- do not rely on it for truly sensitive material. The external converters do not read encrypted files; open in OpenQT, save unencrypted, then convert.

14. Converters

Five command-line tools move documents in and out of OpenQT's format. All are 32-bit DOS4GW builds run from the DOS prompt.

Tool	Direction
oqt2word	OpenQT -> UTF-8 (for Word / LibreOffice)
txt2oqt	UTF-8 or legacy DOS -> OpenQT
txt2rtf	OpenQT (Heb+Eng) -> RTF, keeps bold/underline
oqt2qt	OpenQT -> QText 5.5
qt2oqt	QText 5.5 -> OpenQT (visual -> logical)

oqt2word

```
oqt2word [/R] <input.openqt> <output.utf8>
```

Maps Hebrew to U+05D0-U+05EA, Arabic to the basic block U+0621-U+064A (so the receiving program shapes the letters itself), Arabic-Indic digits to U+0660-U+0669, and writes a UTF-8 BOM. Trilingual files just work -- every byte is classified by range. Add /R for documents written in Russian mode, which routes every high byte through the CP866 table.

txt2oqt

```
txt2oqt <input.txt> <output.openqt> [/V] [/D] [/U] [/R]
```

The inverse of oqt2word. By default it auto-detects UTF-8 via the BOM, reverses the mapping, and converts **bold** / underline markdown markers into format toggles. Flags: /D forces legacy DOS mode, /U forces UTF-8, /V keeps visual order (legacy only), /R targets Russian CP866.

txt2rtf

```
txt2rtf <input.openqt> <output.rtf>
```

Exports a Hebrew + English document to RTF, preserving bold, underline and bold+underline so the styled text opens correctly in Word, LibreOffice and WordPad. Hebrew is emitted as Unicode with right-to-left runs (the renderer does its own bidi). Arabic and Russian are out of scope -- use oqt2word for those. Encrypted files are rejected with guidance.

15. Round-trip workflow

The intended workflow for editing OpenQT documents in modern tools and bringing them back:

- 1. Type and save your document in OpenQT (F2).
- 2. Run `oqt2word doc.txt doc-utf8.txt`.
- 3. Open `doc-utf8.txt` in LibreOffice or Word -- Arabic shapes, Hebrew reads RTL, English stays LTR, all handled by the modern renderer.
- 4. Edit, then save as plain text, UTF-8 encoded.
- 5. Run `txt2oqt doc-utf8.txt doc.txt` to return to OpenQT format.
- 6. Reopen in OpenQT -- same content, re-shaped on screen.

To carry bold/underline through LibreOffice's plain-text export, mark them in the text with **word** and `__word__` markdown -- `txt2oqt` converts those markers back into OpenQT format toggles.

16. Troubleshooting

'OPENQT.HLP not found' when opening Docs

DOSBox's drive cache is stale, or the working directory is wrong. Run 'rescan' in DOSBox after adding the file from the host, and make sure OPENQT.HLP is in the current dir, parent, \OPENQT\ or C:\OPENQT\. Regenerate it with `python3 make_hlp.py`.

OQT 3 shows Hebrew right but Arabic as gibberish (or vice-versa)

Almost always a stale ARABVGA.COM shadowing the new ARABVGA.EXE. DOS resolves .COM before .EXE. Delete ARABVGA.COM, rebuild from `arabvga.c`, and verify ARABVGA /? shows the /P partial-upload help.

The editor opens in Hebrew when I run OQTA

Likely two OPENQT.EXE files exist and DOS picks the older one. Delete the stale copy and rebuild; the current binary is about 75 KB.

Menus look corrupted under OQTA

CP864 reuses the 0xC0-0xDF box-drawing range. The launcher passes /A so dialogs draw with ASCII. If frames show Arabic letters, the /A flag is not reaching the binary -- launch via OQTA, not OPENQT directly.

txt2oqt reads my UTF-8 file as legacy DOS

The file probably lacks a BOM. Save it from LibreOffice with 'Use Unicode (UTF-8)' and 'Include BOM' checked, or pass /U to force UTF-8 mode.

My keys do nothing in DOSBox

Set the DOSBox keyboard to a US layout. OpenQT does its own ASCII -> Hebrew/Arabic mapping, so it expects unmodified ASCII keystrokes.

17. Version history

Version	Highlights
3.9	Hebrew spell check (hspell / aspell-he, follows F4); Copy to Host (Alt+C); Open warns on unsaved changes; Select All works with Copy.
3.8	Paragraph reflow (Block -> Reformat); self-drawn mouse pointer visible in every DOSBox-X mouse mode.
3.7	Mouse support -- clickable menus, click-to-caret, drag-select, wheel scrolling.
3.6	New converter <code>txt2rtf</code> -- Hebrew + English to RTF with bold/underline preserved.
3.5	Smart Paste from the host clipboard (Paste Host / Alt+V).
3.4	Host-assisted Tools: spell check, read aloud, translate, dictate.

3.3	Russian (CP866) input mode and converter /R flag.
3.2	Trilingual editing on one screen; in-editor Docs viewer; full 4-form Arabic shaping.
3.1	First on-screen Arabic shaping pass.
3.0	Arabic input, password encryption, 30,000-line limit.

Earlier versions (1.0-2.x) were Hebrew-only.